🔑 Answer Key — Teacher Copy · Lesson 4-5

MSTAR PRACTICE WORKSHEET

Lesson 4-5: Determine the Whole Given the Part and Percent

Grade 6 Mathematics · Standard 6.AT.4

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

$20 is 50% of what number?

- (A) $10
- (B) $25
- (C) $40
- (D) $70

50% is half, so the part is half the whole. Doubling gives the whole: 20 ÷ 0.5 = 40.

Question 2 SELECTED RESPONSE — CORRECT: A

Which equation says "30 is 25% of some number n"?

- (A) $0.25n = 30$
- (B) $n = 0.25 \times 30$
- (C) $25n = 30$
- (D) $n = 30 + 25$

Percent × whole = part, so $0.25 \times n = 30$.

Question 3 SELECTED RESPONSE — CORRECT: A

$42 is 60% of the original price. Should the original price be more or less than $42?

- A More, because 60% is less than the whole
- B Less, because 60 is a big number
- C Exactly $42
- D You cannot tell

> 60% describes only part of the original price, so the whole original price must be larger than that part.

Question 4 SELECTED RESPONSE — CORRECT: C

75 minutes is 15% of the school day. How long is the school day?

- A 11.25 minutes
- B 90 minutes
- C 500 minutes
- D 1,125 minutes

> Divide the part by the decimal: 75 ÷ 0.15 = 500 minutes.

Question 5 SELECTED RESPONSE — CORRECT: C

A car is now worth $26,600, which is 70% of its purchase price. What was the purchase price?

- A $18,620
- B $26,670
- C $38,000
- D $45,200

> Write 0.7v = 26,600 and divide: v = 26,600 ÷ 0.7 = $38,000.

Question 6 SELECTED RESPONSE — CORRECT: D

Which statement correctly names the pieces of "18 students is 30% of the class"?

- (A) 18 is the whole, 30% is the part
- (B) 30 is the part, 18 is the percent
- (C) The class size is the part
- (D) 18 is the part, 30% is the percent, the class size is the whole

> The known count is the part, the percent describes it, and the missing class size is the whole (100%).

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

Marcus completed 18 math problems, which represents exactly 60% of his homework assignment. How many total problems are on the assignment?

- (A) 24 problems
- (B) 28 problems
- (C) 30 problems
- (D) 36 problems

> Part = Percent × Whole. 18 = 0.60 × Whole. Whole = 18 ÷ 0.60 = 30 problems.

- **A:** 60% of 24 is 14.4. The whole is the part divided by the rate: 18 divided by 0.60.
- **B:** 60% of 28 is 16.8, not the 18 he finished. Divide the part by the rate: 18 divided by 0.60.
- **D:** That doubles the 18, which would make 18 half the assignment rather than 60% of it.

Part B — correct answer: C

Which equation and reasoning justifies the solution in Part A?

- (A) Multiply 18 by 0.60: 18 × 0.60 = 10.8.
- (B) Add 18 and 60: 18 + 60 = 78.
- (C) Since 60% of the whole equals 18, divide the part by the rate: 18 ÷ 0.60 = 30.
- (D) Divide 60 by 18: 60 ÷ 18 = 3.33.

> To find the whole given the part and percent, divide the given part by the percent expressed as a decimal: 18 ÷ 0.60 = 30.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

A theater is at 80% capacity with 240 seats occupied. Select ALL statements that are mathematically true regarding this situation:

- **A** The total seating capacity of the theater is 300 seats (240 ÷ 0.80 = 300).
- **B** There are 60 empty seats in the theater (300 - 240 = 60).
- **C** 10% of the theater capacity is 30 seats (300 ÷ 10 = 30).
- **D** The total capacity is 192 seats (240 × 0.80 = 192).
- **E** A tape diagram with 5 equal units of 60 seats represents the 300 total seats (4 units = 240 = 80%).

> Statements A, B, C, and E are true. Statement D is false because 192 represents 80% of 240 (taking a percent of the part), rather than finding the whole capacity (300).

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A toy store's puzzle stock was 25% 300-piece, 15% 500-piece, and the rest 1,000-piece. During a sale the store sold ALL of its 300-piece and 500-piece puzzles — 120 puzzles in total. How many of each type did the store have before the sale? Explain your reasoning.

> **Model answer:** The 120 puzzles sold were the 300-piece and 500-piece stock together, so they represent 25% + 15% = 40% of the whole stock. That makes 120 the part and 40% the percent, with the total stock unknown: 0.4t = 120, so t = 120 ÷ 0.4 = 300 puzzles in all. Now find each type as a part of that whole: 300-piece = 0.25 × 300 = 75 puzzles, 500-piece = 0.15 × 300 = 45 puzzles, and 1,000-piece = the remaining 60% = 0.6 × 300 = 180 puzzles. Check: 75 + 45 = 120 sold ✓, and 75 + 45 + 180 = 300 ✓.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
